Akoizo Original Practice - Form 1

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is most operational?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Time in seconds between two fixed marks

B. How good the motion is

C. Whether the object seems fast

D. Interesting behavior

Answer: _____

2. (2 points) A fixed +0.5 s offset is…

A. A new independent variable always

B. No error

C. Systematic bias

D. Only random scatter

Answer: _____

3. (2 points) Which is a direct qualitative observation?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Friction definitely increased

B. The hypothesis is true

C. The cause was gravity alone

D. The object wobbled

Answer: _____

4. (2 points) A missing trial should be fabricated to complete the table?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only if software predicts it

B. No

C. Yes

D. Only if the hypothesis is clear

Answer: _____

5. (2 points) A source of error explanation should include…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Only a list of tools

B. Only an apology

C. How it changes the response

D. Only the word “human”

Answer: _____

6. (4 points) Write a directional ramp hypothesis.

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

7. (4 points) Report 2.0 + 3.00 cm.

Answer: _____
_____
_____

8. (4 points) Distinguish SD and standard error.

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

9. (4 points) Why keep raw data?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
_____

10. (4 points) Recommend a targeted improvement for reaction-time variation.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

12. (4 points) Give a targeted follow-up for excessive setup time.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

13. (6 points) Solve this case and explain your reasoning: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

14. (2 points) Define “Control condition” in the context of Experimental Design.

Answer: _____
_____
_____

15. (2 points) Define “Standard of comparison” in the context of Experimental Design.

Answer: _____
_____
_____

16. (2 points) Define “Sample standard deviation” in the context of Experimental Design.

Answer: _____
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_____

17. (2 points) Define “Operational definition” in the context of Experimental Design.

Answer: _____
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_____

18. (2 points) Define “Validity” in the context of Experimental Design.

Answer: _____
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